

ASSIGNMENT NO. 10

Name : Dishitaa Rahul Mahale

Roll no : 23244

Class : SG-10

Batch : G-10

Date of performance : 27-11-2020

Date of completion : 1-12-2020

Title : Factory design pattern

Aim : Design and implement factory design pattern for the given context. Consider car building process, which requires many steps from allocating accessories to final makeup. These steps should be written as methods and should be called while creating an instance of a specific car type hatchback, Sedan, SUV. Test Factory pattern should be implemented.

Objectives : To learn the concept of design pattern.

Theory :

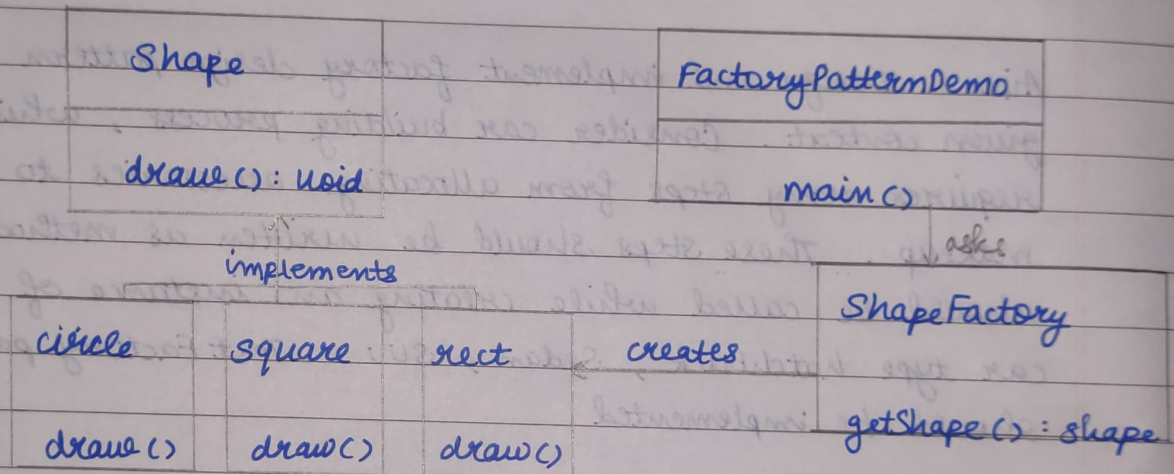
1) Design pattern :

- These are wizards that automate complex, repeatable tasks, and that enable team members with varying level of skills to complete these tasks.
- When extending solutions, you maybe required to repeat design activities multiple times and in a specific order.
- Enables us to define a generic pattern that, when

executed, automates the creation of model objects.

2) Factory design pattern:

Consider a shape interface and concrete classes implementing shape interface. We will have a shape factory as factory class and factoryPatternDemo which will use ShapeFactory to get the type of object it needs.



1) Shape.java

```
public interface Shape {
    void draw();
}
```

2) Rectangle.java

```
public class Rectangle implements Shape {
    @Override
    public void draw() {

```



```
System.out.println (" Draw");
}
```

3) ShapeFactory.java

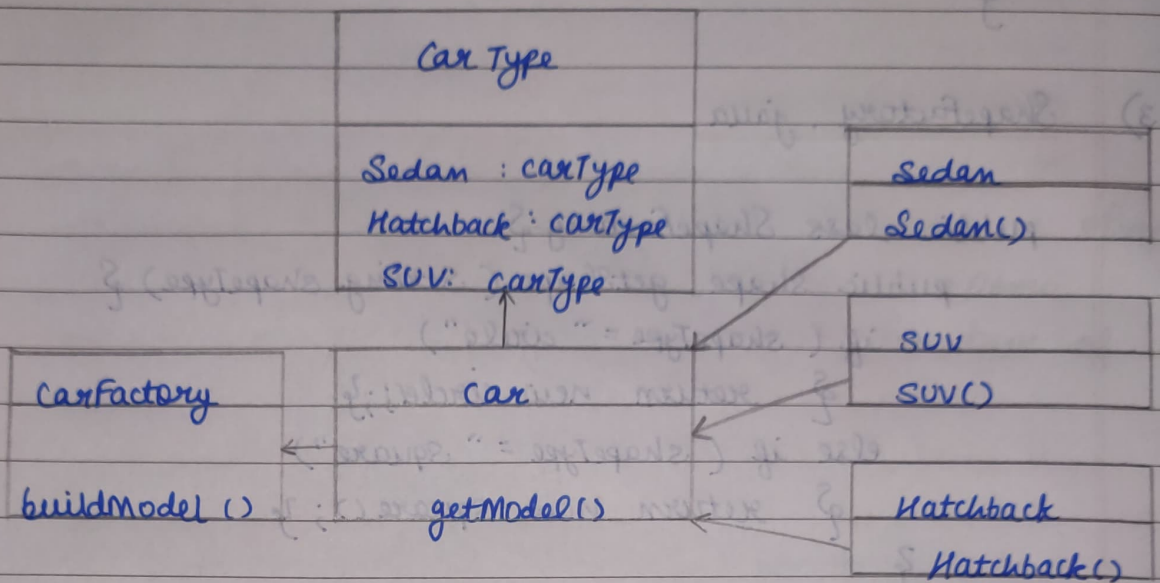
```
public class ShapeFactory {
    public shape getShape (String shapeType) {
        if ( shapeType == " circle")
        {
            return new circle();
        }
        else if ( shapeType == " square")
        {
            return new square();
        }
    }
}
```

4) FactoryPatternDemo.java

```
public class FactoryPatternDemo {
    public static void main (String [] args) {
        ShapeFactory s = new ShapeFactory();
        Shape s1 = ShapeFactory.getShape ("circle");
        Shape s2 = ShapeFactory.getShape ("square");

        s1.draw();
        s2.draw();
    }
}
```

Output: Draw

Class diagram:Advantages:

- The object that we create can be used without duplication of code.
- If we use factory method instead of constructor, it can have different names also.
- It removes the instantiation of implementation classes.
- Makes code less coupled and easy to expand.
- Through inheritance, it provides abstraction between implementation and client classes.